

# Benjamin Boe

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## Education

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| <b>Ph.D.</b> , Astronomy, Institute for Astronomy, University of Hawai'i (IfA/UH)<br>Advisor: Shadia Habbal<br>Thesis: "Solar Wind Formation in the Corona" | August 2020 |
| <b>M.S.</b> , Astronomy, IfA/UH                                                                                                                             | May 2018    |
| <b>B.S.</b> , Physics and Mathematics, University of Puget Sound (UPS)<br>Cum laude and Honors in Physics                                                   | May 2015    |

## Professional Appointments

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|----------------------------------------------------------------------|---------------------------|
| <b>Assistant Professor</b> , Wentworth Institute of Technology (WIT) | August 2023—present       |
| <b>NSF Postdoctoral Fellow</b> , IfA/UH                              | October 2020—March 2023   |
| <b>Faculty Lecturer</b> , Physics & Astronomy, UH                    | August 2021—December 2021 |

## Grants

|                                                                                    |           |           |
|------------------------------------------------------------------------------------|-----------|-----------|
| NSF SHINE Grant (AGS 2501212)                                                      | \$385,959 | 2025–2028 |
| Additional \$182,730 in Collaborative Research grant to Cooper Downs (AGS 2501213) |           |           |
| NSF AGS Postdoctoral Research Fellowship<br>(AGS-PRF 2028173)                      | \$190,000 | 2020–2023 |
| NSO Graduate Fellowship (AURA grant N97991C)                                       | \$122,099 | 2018–2020 |
| NSF Research Experience for Undergraduates (REU)                                   | \$5,000   | 2014      |
| Washington State NASA Space Grant                                                  | \$2,250   | 2013      |

## Publications

### Peer-Reviewed Articles

#### Solar Physics

- Boe, B.**, Habbal, S., Druckmüller, Štarha, P., Štarha, M., Hoderová, J., Constantinou, S., Ayars, E., Casillas, C., *The 2023 Australian Total Solar Eclipse: Line Emission of Fe XIV, Fe X and Fe XI out to 6 solar radii*, ApJ, **Accepted**, [10.48550/arXiv.2512.01068](https://arxiv.org/abs/10.48550/arXiv.2512.01068)
- Benavitz, L., **Boe, B.**, Habbal, S. R. *Total Solar Eclipse White-light Images as a Benchmark for Potential Field Source Surface Coronal Magnetic Field Models: An In-depth Analysis over a Solar Cycle*. ApJ, 974, 178, 2024. [10.3847/1538-4357/ad71c6](https://arxiv.org/abs/10.3847/1538-4357/ad71c6)

3. Zhu, Y., Habbal, S., Ding, A., Yamashiro, B., Landi, E., **Boe, B.**, Constantinou, S., and Nassir, M., *Spectroscopic Observations of the Solar Corona during the 2017 August 21 Total Solar Eclipse: Comparison of Spectral Line Widths and Doppler Shifts between Open and Closed Magnetic Structures*, ApJ, 699, 122, 2024. [10.3847/1538-4357/ad3424](https://doi.org/10.3847/1538-4357/ad3424)
4. **Boe, B.**, Downs, C., Habbal, S., The Solar Minimum Eclipse of 2019 July 2. III. *Inferring the Coronal  $T_e$  with a Radiative Differential Emission Measure Inversion*, ApJ, 951, 55, 2023. [10.3847/1538-4357/acd10b](https://doi.org/10.3847/1538-4357/acd10b)
5. Cooper, J., Habbal, S., **Boe, B.**, Angelopoulos, V., Sibeck, D., Paschalidis, N., Sittler, E., Jian, L., and Killen, R., *Lunar Solar Occultation Explorer (LunaSOX)*, Frontiers in Astronomy and Space Sciences, 10, 1163517, 2023, [10.3389/fspas.2023.1163517](https://doi.org/10.3389/fspas.2023.1163517)
6. **Boe, B.**, Habbal, S., Downs, C., Druckmüller, M., *The Solar Minimum Eclipse of 2019 July 2. II. The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4  $R_\odot$* , ApJ, 935, 173, 2022. [10.3847/1538-4357/ac8101](https://doi.org/10.3847/1538-4357/ac8101)
7. **Boe, B.**, Yamashiro, B., Druckmüller, M., and Habbal, S., *The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse*, ApJ Letters, 914, L39, 2021. [10.3847/2041-8213/ac05ca](https://doi.org/10.3847/2041-8213/ac05ca)
8. Habbal, S., Druckmüller, M., Alzate, N., Ding, A., Johnson, J., Štarha, P., Hoderová, J., **Boe, B.**, Constantinou, S., and Arndt, M., *Identifying the Coronal Source Regions of Solar Wind Streams from Total Solar Eclipse Observations and in situ Measurements Extending Over a Solar Cycle*, ApJ Letters, 911, L4, 2021. [10.3847/2041-8213/abe775](https://doi.org/10.3847/2041-8213/abe775)
9. **Boe, B.**, Habbal, S., Downs, C., Druckmüller, M., *The Color and Brightness of the F-Corona Inferred from the 2019 July 2 Total Solar Eclipse*, ApJ, 912, 44, 2021. [10.3847/1538-4357/abea79](https://doi.org/10.3847/1538-4357/abea79)
10. **Boe, B.**, Habbal, S., Druckmüller, M., *Coronal Magnetic Field Topology From Total Solar Eclipse Observations*, ApJ, 895, 123, 2020. [10.3847/1538-4357/ab8ae6](https://doi.org/10.3847/1538-4357/ab8ae6)
11. **Boe, B.**, Habbal, S., Druckmüller, M., Ding, A., Hodérova, J., and Štarha, P., *CME-induced Thermodynamic Changes in the Corona as Inferred from Fe XI and Fe XIV Emission Observations during the 2017 August 21 Total Solar Eclipse*, ApJ, 888, 100, 2020. [10.3847/1538-4357/ab5e34](https://doi.org/10.3847/1538-4357/ab5e34)
12. **Boe, B.**, Habbal, S., Druckmüller, M., Landi, E., Kourkchi, E., Ding, A., Štarha, P., & Hutton, J., *The First Empirical Determination of the  $Fe^{10+}$  and  $Fe^{13+}$  Freeze-in Distances in the Solar Corona*, ApJ, 859, 155, 2018. [10.3847/1538-4357/aabfb7](https://doi.org/10.3847/1538-4357/aabfb7)

### Long Period Comets

1. **Boe, B.**, Jedicke, R., Meech, K., Wiegert, P., Weryk, R., et al., *The orbit and size-frequency distribution of long period comets observed by Pan-STARRS1*, Icarus, 333, 252, 2019. [10.1016/j.icarus.2019.05.034](https://doi.org/10.1016/j.icarus.2019.05.034)

### Galaxy Evolution

1. Shields, D., **Boe, B.**, Pfountz, C., Davis, B., et al., *Spirality: A novel way to measure spiral arm pitch angle*, Galaxies, 10, 100, 2022. [10.3390/galaxies10050100](https://doi.org/10.3390/galaxies10050100)

### Musical Acoustics

1. **Boe, B.**, and Randy Worland, *Experimental and numerical analysis of the effect of length in musical drumhead coupling*, Proceedings of Meetings on Acoustics, 20, 1, 2014. [10.1121/1.4884779](https://doi.org/10.1121/1.4884779)

### Non-Peer-Reviewed

White Papers for the 2024 Solar and Space Physics (Heliophysics) Decadal Survey

1. **Boe, B.**, Habbal, S., and Ding, A., *The Need for a New Generation of Space-based Visible and Near-IR Emission Line Observations of the Corona*, 2023, [10.3847/25c2cfef.fb89d0ac](https://doi.org/10.3847/25c2cfef.fb89d0ac)
2. Habbal, S., **Boe, B.**, Haggerty, C., and Ding, A., *A Total Solar Eclipse Earth-Based Mission: Multi-wavelength Observations from Land, Sea and Air to Probe the Critical middle Corona*, 2023, [10.3847/25c2cfef.70c8c8e3](https://doi.org/10.3847/25c2cfef.70c8c8e3)
3. Cooper, J., Habbal, S., **Boe, B.**, et al., *Heliophysics from, by, and of the Moon in the Age of Artemis: Lunar Solar Occultation Explorer (LunaSOX)*, 2023, [10.3847/25c2cfef.beac35c4](https://doi.org/10.3847/25c2cfef.beac35c4)

### Talks, Presentations, and Workshops

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#### Workshops

1. Invited Workshop Participation, “Eclipses and Beyond: Unveiling the Mysteries of the Sun's Visible Corona”, organized and sponsored by the International Space Science Institute (ISSI), Bern, Switzerland, 2025.

#### Colloquia

1. “Coronal science with Narrowband and White-light imaging from Eclipses”, Invited Colloquium. University of Massachusetts Lowell, virtual, 2025
2. “The Scientific Value of Total Solar Eclipses”, Invited Colloquium. California State University, virtual, 2022
3. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. University of Puget Sound, Tacoma, WA, 2020
3. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. Bridgewater State University, virtual, USA, 2020

4. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. Aberystwyth University, Aberystwyth, Wales, 2019
5. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Colloquium. Institut d’Astrophysique Spatiale, Orsay, France, 2019
6. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Colloquium. Brno University of Technology, Brno, Czech Republic, 2019
7. “Studying Solar Coronal Physics with Total Solar Eclipses”, Invited Colloquium. University of Puget Sound, Tacoma, WA, 2019
8. “Testing Solar System Formation With Manx Comets”, Colloquium. UH NASA Astrobiology Institute Seminar, University of Hawai’i, 2016

### Talks and Posters

1. “Coronal science with Narrowband and White-light imaging from Eclipses”, Invited Talk. Center for Astrophysics | Harvard & Smithsonian (CfA), Cambridge, MA, 2025
2. “Advances in Total Solar Eclipse Narrowband Observations: Introducing a Radiative Differential Emission Measure”, Talk. American Geophysical Union (AGU) Meeting, District of Columbia, 2024
3. “Insights from a Total Solar Eclipse Research Expedition: Educating Students and Engaging the Public”, Talk. American Association of Physics Teachers (AAPT) meeting, Boston, MA, 2024
4. “The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4 Rs”, Talk. Triennial Earth-Sun Summit, Bellevue, WA, 2022
5. “The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4 Rs”, Poster. Solar Heliospheric and Interplanetary Environment (SHINE) Workshop, Honolulu, HI, 2022
6. “The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse”, Poster. AGU Meeting, hybrid (virtual), 2021
7. “The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse”, Poster. SHINE Workshop, virtual, 2021
8. “Coronal Magnetic Field Topology From Total Solar Eclipse Observations”, iPoster. American Astronomical Society (AAS) Solar Physics Division Meeting, virtual, 2020
9. “Thermodynamic Changes in the Corona during the 2017 August 21 Total Solar Eclipse”, Talk. AAS Meeting, Honolulu, HI, 2020
10. “CME Induced Thermodynamic Changes in the Corona as Inferred from Fe XI and Fe XIV Emission Observations during the 2017 August 21 Total Solar Eclipse”, Poster. AGU Meeting, San Francisco, CA, 2019
11. “The Orbit and Size-Frequency Distribution of Long Period Comets Observed by PS1”, Talk. EPSC-DPS Joint Meeting, Geneva, Switzerland, 2019
12. “CME Induced Thermodynamic Changes in the Corona as Inferred by Fe XI and Fe XIV Emission Observations from the 2017 August 21 Total Solar Eclipse”, Poster. SHINE Workshop, Boulder, CO, 2019

13. “First Empirical Determination of the Fe 10+ and Fe 13+ Freeze-in Distances in the Solar Corona”, Poster. SHINE Workshop, Cocoa Beach, FL, 2018
14. “First Empirical Determination of the Fe 10+ and Fe 13+ Freeze-in Distances in the Solar Corona”, Talk. Asia Oceania Geosciences Society Meeting, Honolulu, HI, 2018
15. “Breaking the Wall of Solar Wind Formation”, Talk. Falling Walls Lab, UH, 2017
16. “Testing Solar System Formation Models Using Pan-STARRS1 Detections of Nearly Inactive Manx Comets”, Talk. Division of Planetary Sciences Meeting, Pasadena, CA, 2016
17. “Pitch Angle Survey of GOODS Spiral Galaxies”, Poster. AAS Meeting, Seattle, WA, 2015
18. “Pitch Angle Survey of GOODS Spiral Galaxies”, Poster. Murdock College Science Research Program (MCSR) Conference, Vancouver, WA, 2014
19. “Experimental and numerical analysis of the effect of length in musical drumhead coupling”, Poster. Acoustical Society of America Meeting, San Francisco, CA, 2013
20. “Coupled Vibrations Between Musical Drumheads”, Talk. MCSR Conference, Vancouver, WA, 2013

### **Observing Experience and Telescope Time**

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Daniel K Inouye Solar Telescope (DKIST)

Three successful proposals as PI (Cycles 2, 3 and 4) and one as Co-I (Cycle 4)

Total Solar Eclipse Expeditions

Preparation in lab and on-site operation, alignment, and calibration of instrumentation

|                      |                    |
|----------------------|--------------------|
| Kerrville, Texas     | April 8th, 2024    |
| Exmouth, Australia   | April 20th, 2023   |
| Antarctic Ocean      | December 4th, 2021 |
| Vicuña, Chile        | July 2nd, 2019     |
| Alliance, Nebraska   | August 21st, 2017  |
| Halmahera, Indonesia | March 9th, 2016    |

Maunakea Observing

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|------------------------------|---------------------------|
| Subaru HSC, summit observing | January 3rd and 4th, 2016 |
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### **Student Projects**

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Advised 6 undergraduate students in academic research

- Three are now in graduate school
- One paper published in ApJ (Benavitz et al. 2024, see above)
- Three publications currently in prep
- Two were senior projects, one internally funded summer project, one NSF REU, and two Co-ops (semester long internship)

Advised one graduate student

- Student of my former PhD advisor (Shadia Habbal)
- Provided DKIST data for their thesis, and meet with them regularly
- Publication in prep

## Teaching Experience

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### Astronomy Courses

**Foundations of Astrophysics I: The Solar System** (ASTR 214) — 1 section at UH  
Calculus-based introduction to orbital mechanics, tidal interactions, planetary interiors and atmospheres, small bodies, hydrostatic/thermal equilibrium, and solar & heliospheric physics.

**Introduction to Astronomy** (ASTR 110; PHYS 2000) — 3 UH lecture; 2 WIT lecture+lab  
Survey of sky motion, astronomical methods and instrumentation, planetary and stellar physics, galaxies, and cosmology

**Introduction to Astronomy Lab** (ASTR 110L) — 3 sections at UH  
Hands-on observational laboratory including nighttime telescope operation and data-driven activities.

**Space Exploration** (PHYS 2300) — 1 section at WIT with lab  
Original course integrating algebra-based physics with spaceflight history, emphasizing spacecraft dynamics and mission design.

### Physics Courses

**Modern Physics** (PHYS 3100) — 1 section at WIT  
Special relativity, blackbody radiation, atomic structure, photoelectric effect, scattering processes, introductory quantum mechanics.

**Engineering Physics 1** (PHYS 1250) — 6 sections at WIT with labs  
Mechanics: kinematics, Newton's laws, work–energy, momentum, rotational dynamics.

**Engineering Physics 2** (PHYS 1750) — 4 sections at WIT with labs  
Electric and magnetic fields, DC circuits, and geometric optics

### Service

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|-----------------------------------------------------------|--------------|
| WIT Liaison, Massachusetts Space Grant Consortium (NASA)  | 2024-present |
| Astronomy Club Faculty Advisor, Colleges of the Fenway    | 2023-present |
| Course Coordinator for Physics 1250 and 1750, 3 semesters | 2025-present |

### Referee

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|---------------------------------------|-----------------------------------------|
| The Astrophysical Journal, 9 articles | NSF Grant Proposals, 2 panels, 2 ad hoc |
| Solar Physics, 5 articles             | DKIST Review, 1 panel                   |

## Selected Press and Outreach

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Forbes article about Boe et al. (2020b), [link](#)  
NASA article about Habbal et al. (2021), with interview, [link](#)  
The Guardian article about 2023 eclipse expedition, [link](#)  
WABI Bangor Maine Local News, interview, March 22nd, 2024, [link](#)  
Boston Museum of Science, public talk, March 26th, 2024  
Kerrville, Texas, public talk at eclipse festival, April 7th 2024  
Texas Standard (public radio), live on-air interview, April 8th, 2024, [link](#)  
Astronomy club, with my mentorship, hosted eclipse viewing event on WIT campus, April 8th, 2024  
NASA's Curious Universe Podcast, May 3rd, 2024, interview, [link](#)

## Honors, Awards, and Scholarships

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|--------------------------------------------|----------|-----------|
| Sigma Pi Sigma Physics Honor Society       |          | 2015      |
| University of Puget Sound Dean's list      |          | 2014      |
| Raymond & Olive Seward Scholarship         | \$3,500  | 2014      |
| Acoustical Society of America poster award | \$500    | 2013      |
| John Van Zytveld Award in Physical Science | \$2,000  | 2013      |
| Thomas & Hilda Jack Scholarship            | \$2,750  | 2013      |
| Puget Sound President's Scholarship        | \$64,000 | 2011–2015 |
| Kilworth Foundation Scholarship            | \$2,500  | 2011      |